Geometry, Entropy, and Operator Learning with Tunable Kernels

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Abstract

We study a two-dial family of kernels

$$k_{\theta,\sigma}(y \mid x) \propto w_\theta\big(\|x-y\|\big)\; g_\sigma(y-x),$$

whose product form covers any radial weight w_θ and centred residual density g_σ that satisfy minimal smoothness and moment conditions. To give the theory a concrete footing—and to obtain closed-form bias constants in our finite-sample analysis—we instantiate this family with a data-adaptive *local-linear* construction, but all results hold for the broader class.

For every observable f we compute the one-step cumulant-generating function (CGF) and show that its quadratic coefficient $\mathcal{I}_{\theta,\sigma}[f]$ serves simultaneously as (i) a *Fisher load*, (ii) $\frac{2}{\theta^2}$ times the synthetic Ricci curvature $\Gamma_2^{(\theta,\sigma)}(f)$, and (iii) $-2\,\partial_\sigma$ of the one-step Shannon entropy. This *Fisher–curvature–entropy bridge* yields a single quadratic objective whose minimiser in the dial plane, $(\theta^\star,\sigma^\star)$—the *little-rose*—optimises curvature bias, entropy production, and operator-learning risk at once. We prove

$$\nabla_{(\theta,\sigma)}\mathrm{KL}\big(\mu_1 \,\|\, k_{\theta,\sigma}\#\mu_0\big) \;=\; -\tfrac{1}{2}\,\nabla_{(\theta,\sigma)}\mathcal{I}_{\theta,\sigma},$$

so steepest KL descent converges to the little-rose with no hyper-parameters. A data-driven Goldenshluger–Lepski selector recovers $(\hat\theta_\star,\hat\sigma_\star)$ from finite samples, and we give non-asymptotic risk bounds for both Perron–Frobenius and Koopman estimates.

The resulting dial matrices $(\Theta_{ij},\Sigma_{ij})$ form a directed, doubly-weighted graph, opening avenues toward discrete curvature flows, entropy production on cycles, and multiparameter persistent homology—highlighting a promising intersection of statistical dynamics, information geometry, and topological data analysis.

1 Introduction

1.1 Why tune local kernels?

1.2 Key contributions

- Closed-form bridge: CGF curvature → Fisher load → curvature & entropy.
- Single KL optimisation (little-rose) jointly tunes **two** dials (θ,σ).
- Finite-sample Goldenshluger–Lepski selector with risk guarantees.

1.3 Organisation of the paper

2 Rose kernels and dials

2.1 Product-form kernels and minimal assumptions (K0–K3)

$$z := y - x, \qquad r := \|z\|.$$

(W1) (**Radial weight**) $w : [0, \infty) \to [0, \infty)$ is C^1, radial, $\int_{\mathbb{R}^d} w(\|z\|)\, dz = 1$, $\int \|z\|^2 w(\|z\|)\, dz < \infty$, and $w \in L^2(\mathbb{R}^d)$.

(W2) (**Centred residual**) $g : \mathbb{R}^d \to [0, \infty)$ is C^1, $\int g(z)\, dz = 1$, $\int z\, g(z)\, dz = 0$, $\int zz^\top g(z)\, dz < \infty$, and $\nabla g \in L^2(\mathbb{R}^d)$.

Definition 2.1 (Rose kernel)**.** For $(\theta, \sigma) \in (0, \infty)^2$ set

$$w_\theta(r) := \theta^{-d} w\big(r/\theta\big), \quad g_\sigma(z) := \sigma^{-d} g\big(z/\sigma\big), \quad k_{\theta,\sigma}(y \mid x) := \frac{w_\theta(\|x - y\|)\, g_\sigma(y - x)}{\displaystyle\int_X w_\theta(\|x - y'\|)\, g_\sigma(y' - x)\, dy'}$$

(K1) (**Markov**) $k_{\theta,\sigma}(y \mid x) \geq 0$ and $\int_X k_{\theta,\sigma}(y \mid x)\, dy = 1$.

(K2) (**Score in L^2**) $S_{\theta,\sigma}(y \mid x) := \nabla_y \log k_{\theta,\sigma}(y \mid x) \in L^2(dy\, dx)$.

(K3) (**Second-moment scaling**) $\displaystyle\int_X \|y - x\|^2 k_{\theta,\sigma}(y \mid x)\, dy = O(\theta^2)$ as $\theta \downarrow 0$.

(K4) (**C^1 in dials**) $\partial_\theta \log k_{\theta,\sigma}, \partial_\sigma \log k_{\theta,\sigma} \in L^2(dy\, dx)$.

2.2 Local–linear residual construction (canonical example)

2.3 Scaled kernels and notation for the rest of the paper

3 Population theory

(K0)–(K2) as in Sec. 3.0 — Markov, score L^2, second-moment scaling.

3.1 CGF ⇒ Fisher ⇒ Landau

Definition 3.1 (One–step CGF)**.**

Proposition 3.1 (CGF–Fisher correspondence)**.**

Remark 3.1 (Legendre dual = Landau stiffness). A two–line box reminding readers that the Legendre transform of Λ yields a quadratic Landau free energy.

3.2 Fisher–curvature–entropy bridge

Lemma 3.1 (Curvature identity)**.**

Lemma 3.2 (Discrete de Bruijn).

Theorem 3.1 (Fisher–curvature–entropy bridge).

3.3 Introducing the dial plane

(K0) C^1 regularity in dials. The maps $(\theta,\sigma)\mapsto \log k_{\theta,\sigma}$ are C^1 ...

3.4 KL gradient and the *little-rose* optimiser

Theorem 3.2 (KL gradient).

Definition 3.2 (Little-rose).

Corollary 3.1 (Unified optimality).

3.5 Why it matters (P1–P5 summary)

4 Finite-sample dial selection

4.1 Goldenshluger–Lepski selector

4.2 Risk bounds for PF and Koopman estimates

Theorem 4.1 (GL selector, risk bound).

4.3 Illustrative experiment

5 Discussion and Outlook

5.1 Key implications

(I1) Unified diagnostic of fluctuation strength
The Fisher load $\mathcal{I}_{\theta,\sigma}$ is *simultaneously* the variance coefficient, the synthetic–curvature prefactor, and the entropy–production rate. A single dial pair (θ,σ) governs them all.

(I2) Automatic bias–variance balance
Steepest descent of $\mathrm{KL}(\mu_1 \parallel k_{\theta,\sigma}\#\mu_0)$ is steepest *ascent* of $\mathcal{I}_{\theta,\sigma}$, so the statistically most informative kernel is found with no cross-validation.

(I3) One-step Schrödinger bridge
The little-rose kernel realises the discrete entropic transport $\mu_0\to\mu_1$, linking operator learning to optimal-transport theory.

(I4) Landau lens on data
The Legendre transform of the CGF equips any observable with a Landau-type free-energy landscape whose stiffness is data-computable.

5.2 Future directions

(F1) Directed discrete curvature
Dial matrices $(\Theta_{ij}, \Sigma_{ij})$ define a doubly-weighted digraph, opening analytic work on Ollivier/Bakry–Émery curvature for non-reversible kernels.

(F2) Multiparameter persistent homology
The two weights furnish a natural bifiltration, inviting multi-parameter TDA of dynamical data sets.

5.3 Limitations

5.4 Conclusion

A Proof details for Sec. 3

B GL selector derivations

C Additional experiments

D Dial-graph topology (exploratory)